

INTRODUCTION

This project analyses customer purchase behaviour using the [“Online Retail.xlsx”](#) dataset obtained from kaggle.com. It contains transactions records from a UK-based online retail store covering the period from 1 December 2010 to 9 December 2011. The dataset include information on Invoices, products, quantities purchased, transaction dates, prices, customers and countries.

The analysis focuses on understanding customer-level purchasing behaviour, identifying patterns in sales, segmenting customers according to their historical purchasing characteristics and using the information to predict whether they are likely to become inactive. In this project, customer inactivity is defined as making no purchase in the 90-day period prior to an observation date. The resulting analysis can be used to support business decisions relating to customer retention, customer prioritization and purchasing behaviour.

BUSINESS PROBLEM AND OBJECTIVES

The purpose of this project is to use historical transaction data to understand customer and sales behaviour and to develop a predictive approach for identifying customers who may become inactive.

The analysis addresses the following questions:

- Which Countries generate the highest net spend?
- Which Countries generate the lowest net spend?
- Which products are returned most frequently and in the greatest quantities?
- What percent of revenue is generated by the top 20% of customers by spending?
- Which products are purchased most frequently and in the greatest quantities?
- How does revenue vary across quarters?
- Which customer segments generate the greatest proportion of revenue?
- Can historical data be used to predict customer inactivity?

DATASET STRUCTURE AND PREPARATION

The original dataset contains 541 909 transaction records, with 8 variables: *InvoiceNo*, *StockCode*, *Description*, *Quantity*, *InvoiceDate*, *UnitPrice*, *CustomerID* and, *Country*.

Returned products are represented by negative values in the *Quantity* Column while positive values represent purchased items. The analysis therefore separates sales and returns while examining purchasing and return behaviour.

A new variable, *TransactionValue* was created by multiplying *UnitPrice* by *Quantity*
$$\text{TransactionValue} = \text{UnitPrice} \times \text{Quantity}$$

Transactions without *CustomerID*'s were removed because the project contains customer level analysis and these records cannot be reliably attributed to an individual customer. Missing product descriptions were also replaced with “No Description”.

After removing transactions without a *CustomerID*, 406829 records remained.

EXPLORATORY DATA ANALYSIS

Exploratory data analysis was carried out to identify major patterns in customer purchasing behaviour, product demand, geographical performance and revenue over time. Transactions involving positive quantities were treated as sales while negative quantities represented returns

Country Performance

Business Question

Which country generated the highest net spend

Method

Net spend was calculated by aggregating *TransactionValue* by country. The analysis considered the total value generated by each country, while taking account of negative transaction values associated with returns.

Results

Highest Net Spend

United Kingdom	7308391.554
Netherlands	285446.340
EIRE	265545.900
Germany	228867.140
France	209024.050
Australia	138521.310

Interpretation

This indicates that the business is heavily concentrated in the UK market, the presence of substantial sales in other European and international markets suggests opportunities for further geographical expansion and customer development.

Product Purchasing Behaviour

Business Questions

- Which products are returned most frequently?
- Which products have the largest quantities returned?

Method

Returned products were identified using transactions with negative quantities. Return frequency was measured using the number of distinct invoices associated with returned products.

Results

Most returned products by quantity

Description	
PAPER CRAFT , LITTLE BIRDIE	80995

Most Returned Products by frequency	
Description	
REGENCY CAKESTAND 3 TIER	180
Manual	154
POSTAGE	95
JAM MAKING SET WITH JARS	86
SET OF 3 CAKE TINS PANTRY DESIGN	

Products with high return frequency would warrant further investigation to determine whether returns are due to poor product quality, customer expectations or other operational factors

Business Question

Method

Result

	InvoiceDate	TotalRevenue	OrderCount
0	2010Q4	554604.020	1708
1	2011Q1	1491585.140	4057
2	2011Q2	1682312.091	4940
3	2011Q3	2122046.853	5215
4	2011Q4	2449517.710	6270

Revenue increased throughout the observed period rising from approximately 554604 to 2449517 which represented a substantial increase across the measured period. The upward trend suggests increasing sales activities across the study period. Further analysis would be required to determine the extent to which the increase is driven by seasonal variations, greater purchasing frequency, larger orders or customer acquisition

Customer Revenue Concentration

Business Question

What percentage of revenue is generated by the top 20% of customers

Method

Customers were ranked according to their total monetary value and the highest spending 20% were compared with the remaining customers to determine the concentration of revenue

Results

The top 20% of customers were responsible for an average of 73.85% of total revenue

Intepretation

This indicates that revenue is highly concentrated at a relatively small proportion of customers, These high-value customers should receive particular attention in customer retention strategies as losing a relatively small number of high value customers could have a disproportionate effect on revenue

RFM ANALYSIS

Purpose

Recency, Frequency and Monetary value analysis was conducted to further understand customer behaviour.

Recency measures how recently a customer made a purchase, **Frequency** measures the number of purchases made **Monetary** value measures the total value of a customer's transaction.

Method

Recency was calculated as the number of days between the last transaction and the customer's most recent purchase. Frequency was calculated as the number of distinct invoices associated with a customer within the observation window while monetary value was calculated as the customer's total transaction value.

Customers were then assigned RFM scores using quintiles and were further classified into segments based on combinations of their RFM scores, customers were classified into segments including champions, Loyal High-value customers, potential loyalists, At Risk High value, at risk low value, Hibernating/Average and Lost.

Results

The RFM segmentation produced the following distribution:

Segment	Customers	Customer %	Revenue %
Champions	976	22.3%	68.1%
Loyal High-value Customers	420	9.6%	13.2%

Segment	Customers	Customer %	Revenue %
Hibernating / Average	1,493	34.1%	9.2%
Lost	1,058	24.2%	4.8%
Potential Loyalists	306	7.0%	3.0%
At Risk High Value	80	1.8%	1.5%
At risk low-value	39	0.9%	0.1%

Intepretation

The RFM analysis shows that a relatively small group of highly engaged customers contributes a disproportionately large share of revenue business (Revenue is concentrated at the top)

This suggests that customer retention strategies should not treat all customers equally. Particular attention should be given to high value customers whose purchasing behaviour begin to indicate declining engagement.

PARAETO ANALYSIS

Business Question

How concentrated is revenue among the highest spending customers?

Method

Customers were ranked from highest to lowest according to monetary value. Cummulative revenue and customer percentages were then calculated to determine the extent to which revenue is concentrated across percentages (20% in the case of the paraeto principle)

Results

The dataset contains 4,372 unique customers with a total revenue of about 8.3 million.

The highest-spending 20% of customers representing 875 customers were responsible for 73.85% of total revenue. Furthermore, approximately 26.78% of customers accounted for 80% of total revenue

Intpretation

The results show a strong concentration of revenue among higher value customers. Although the conventional 80/20 relationship is not exact in this dataset, the top 20% of customers still generated more than 3 quaters of the total revenue.

This highlights the importance of customer retention strategies especially among high value customers.

CUSTOMER CHURN PREDICTION

Churn Definition

For this analysis, a customer is classified to be churned or inactive if they make no purchase during the 90 day window following an observation date.

The model does not attempt to determine permanent customer loss instead it estimates the probability that an active customer would become inactive over the next 90 days.

This distinction is important as a customer who is inactive for 90 days may return to purchase at a later date.

CHURN DATASET CONSTRUCTION

Method

Rather than creating a single customer-level snapshot, multiple monthly observation dates were used, with customer data available to each respective date being used to calculate predictive features.

The subsequent 90-day period was then examined to determine whether each customer made a purchase. Customers who made no purchase during that period were assigned a churn value of 1, while customers who made at least one purchase were assigned a value of 0.

Seven monthly observation dates were available for the modelling exercise.

After constructing the snapshots, customers with fewer than two historical purchases were removed to ensure that behavioural features such as purchase intervals were based on sufficient purchasing history. This resulted in **9,386 customer-period observations**.

FEATURE ENGINEERING

The predictive model was constructed using behavioural features intended to capture customer engagement, value, purchasing consistency and returns

Feature group	Features
Customer value	Monetary, Average Order Value
Purchase engagement	Recency, Frequency, Orders Last 30 Days
Recent spending	Spend Last 30 Days, Spend Previous 30 Days, Spend Change
Product behaviour	Average Basket Size, Unique Products
Purchase consistency	Average Days Between Orders, Standard Deviation of Days Between Orders
Returns	Return Count, Returned Quantity, Returned Value, Return Rate

TEMPORAL TRAIN/TEST STRATEGY

Since the data was time sensitive, a chronological train/test split was used in order to better reflect the intended business application of the model.

BASELINE MODEL

A recency only baseline was created to determine whether the additional behavioural features provided predictive value beyond one of the strongest traditional churn indicators

The baseline produced:

Metric	Recency Baseline
ROC-AUC	0.6472
PR-AUC	0.4683

LIGHTGBM MODEL

Method

A lightGBM binary classifier was used to estimate the probability that a customer would become inactive during the following 90 days.

The model was trained using 19 behavioural features. The implementation used a learning rate of 0.02, 300 estimators, 31 leaves, a minimum child sample size of 20 and L1/L2 regularisation.

Hyperparameter tuning was not carried out because the primary objective of the model was to establish a robust temporal modelling framework and compare the behavioural model against the simpler baseline.

MODEL EVALUATION

The final LightGBM model achieved the following results in the temporally held out test set data

Metric	LightGBM
ROC-AUC	0.7759
PR-AUC	0.5925
Precision	0.5850
Recall	0.5405
F1-score	0.5619

The confusion matrix was

	Predicted No Churn	Predicted Churn
Actual No Churn	2,099	454
Actual Churn	544	640

Comparison with the baseline

Model	ROC-AUC	PR-AUC
Recency baseline	0.6472	0.4683
Full LightGBM	0.7759	0.5925

Interpretation

The improvement over the baseline suggests that inactivity cannot be explained by recency alone. Additional information provided by spending behaviour, purchase frequency, product diversity, purchasing consistency and returns provide useful predictive insights,

The recall of 0.5405 shows that the model correctly identified ~54% of the customers classified as inactive within the test period. The precision of 0.585 indicates that approximately 59% of customers predicted as inactive were actually classified as inactive.

Due to the results, this model is to be better viewed as a risk ranking tool rather than a perfect classifier.

FEATURE IMPORTANCE

The ten most important features were:

Rank	Feature
1	Average Basket Size
2	Standard Deviation of Days Between Orders
3	Purchased Quantity
4	Average Order Value
5	Average Days Between Orders
6	Unique Products
7	Monetary
8	Return Rate
9	Recency
10	Returned Value

Intepretation

The feature importance results suggest that purchasing consistency and basket behaviour are particularly useful for distinguishing customers who later become inactive. Measures such as the variability of time between purchases, basket sizes, average order value and product diversity were more important in the model than recency alone.

BUSINESS RECOMMENDATIONS

The analysis provides several potential areas for business action.

Customer retention

The strong revenue concentration among high-value customers indicates that retention should prioritise customers who contribute substantial revenue. The combination of RFM segmentation and churn risk can be used to distinguish valuable customers who are also showing signs of declining engagement.

Targeted retention campaigns

Rather than applying the same retention strategy to all customers, the business could use predicted churn probability to prioritise customers for intervention. High-value customers with elevated churn risk should receive greater attention than low-value customers with similar risk.

Monitor purchasing deterioration

The predictive model gives significant importance to behavioural measures such as purchase intervals, basket size, monetary value and product diversity. Monitoring changes in these behaviours could help identify customers whose engagement is weakening before they become fully inactive.

Investigate returns

The analysis identified substantial return activity for certain products, while Return Rate also contributed to the churn model. Products and customers with unusually high return behaviour could therefore be investigated to identify possible quality, fulfilment or expectation-related issues.

Inventory planning

The product analysis identified products with high purchasing quantities and high purchase frequency. These measures can support inventory planning, although purchasing frequency, quantity and revenue should be considered together rather than relying on a single measure.

LIMITATIONS

Several limitations should be considered when interpreting the findings.

First, the dataset covers a relatively short period, resulting in only **seven monthly observation dates** for the churn modelling exercise. This limits the number of independent temporal periods available for validation.

Second, the 90-day churn definition represents **customer inactivity rather than permanent churn**. A customer classified as inactive may return to purchase later.

Third, the monthly customer snapshots have overlapping 90-day prediction windows. This means that customer-period observations are not fully independent.

Fourth, the dataset contains transaction information but does not provide additional information such as marketing exposure, customer demographics, customer service interactions or reasons for returns. Consequently, the model is limited to behavioural patterns observable in transaction history.

Finally, the model achieved useful but imperfect predictive performance. A ROC-AUC of approximately 0.78 indicates meaningful discrimination, but the model should not be treated as perfectly identifying future inactive customers.

CONCLUSION

This project analysed customer purchasing behaviour using transaction data from an online retail business. The exploratory analysis demonstrated substantial geographical, product and temporal differences in business activity, while the RFM analysis showed that customer value is highly concentrated within a relatively small proportion of the customer base.

The Pareto analysis further demonstrated this concentration, with the top 20% of customers generating approximately 73.85% of revenue. This highlights the importance of understanding and retaining valuable customers.

A 90-day inactivity framework was subsequently developed to predict customers who may become inactive. A LightGBM model using historical purchasing, spending, product, interval and return

features achieved a ROC-AUC of **0.7759** and a PR-AUC of **0.5925**, substantially outperforming the recency-only baseline.

Overall, the results suggest that customer inactivity can be predicted to a useful degree using transaction-based behavioural features. The model therefore provides a basis for prioritising customers for retention activities, while the RFM and exploratory analyses provide additional context for deciding how those interventions should be targeted.